

Building CI-CD Pipeline 2 (CI)

Github Actions For CI

For the CI stage, GitHub Actions installs dependencies, runs security checks, builds the frontend, and validates that backend and frontend Docker images can be built successfully. The images are not pushed to a container registry yet.



GitHub Action CI

Create GitHub Action CI File

1. Create GitHub Action CI File within a `.github/` folder within the local machine - `.github/workflows/ci.yaml`



2. Make sure the `ci.yaml` have the relevant content.

```
# Workflow name displayed in GitHub Actions
name: SITWallet CI

# Trigger workflow when:
# 1. Code is pushed to main branch
# 2. Pull Request is opened/updated against main branch
on:
  push:
    branches: ["main"]
  pull_request:
    branches: ["main"]

jobs:

  # =====
  # Backend Continuous Integration
  # =====
  backend-ci:

    # Job name displayed in GitHub Actions UI
    name: Backend CI

    # GitHub-hosted runner (Ubuntu Linux VM)
    runs-on: ubuntu-latest

    # Set default working directory for all run commands
    defaults:
      run:
        working-directory: backend

    steps:

      # Step 1: Download repository source code
      - name: Checkout repository
        uses: actions/checkout@v4

      # Step 2: Install Node.js environment
      - name: Setup Node.js
        uses: actions/setup-node@v4
        with:

          # Use Node.js version 20
```

```

node-version: 20

# Enable npm dependency caching
cache: npm

# Cache based on backend package-lock.json
cache-dependency-path: backend/package-lock.json

# Step 3: Install backend dependencies
# npm ci ensures clean installation based on package-lock.json
- name: Install backend dependencies
  run: npm ci

# Step 4: Check for high severity vulnerabilities
# Workflow fails if high severity issues are found
- name: Run backend audit
  run: npm audit --audit-level=high

# Step 5: Verify Docker image can be built successfully
# No image is pushed yet; this is only a build validation
- name: Build backend Docker image
  run: docker build -t sitwallet-backend ./

# =====
# Frontend Continuous Integration
# =====
frontend-ci:

# Job name displayed in GitHub Actions UI
name: Frontend CI

# GitHub-hosted Ubuntu runner
runs-on: ubuntu-latest

# Set frontend as default working directory
defaults:
  run:
    working-directory: frontend

steps:

# Step 1: Download repository source code
- name: Checkout repository
  uses: actions/checkout@v4

# Step 2: Install Node.js environment
- name: Setup Node.js
  uses: actions/setup-node@v4
  with:

    # Use Node.js version 22
    node-version: 22

    # Enable npm dependency caching
    cache: npm

    # Cache based on frontend package-lock.json
    cache-dependency-path: frontend/package-lock.json

# Step 3: Install frontend dependencies
# Ensures consistent installation from lock file
- name: Install frontend dependencies
  run: npm ci

# Step 4: Build frontend application
# Verifies React/Vue/Angular build succeeds
- name: Build frontend
  run: npm run build

# Step 5: Verify frontend Docker image builds successfully
# Ensures Dockerfile is valid and application can be containerized
- name: Build frontend Docker image
  run: docker build -t sitwallet-frontend ./

```

Clone GitHub Repository

GitHub recognises both `yaml` and `yml` as their CI file.

3. Go to the `2216-ssd-10-main` folder and `git clone https://github.com/koksenglim0/2216-ssd-10-main.git`

Once the command is being executed, it means repository has been cloned into the local Repo folder **inside a folder called** `2216-ssd-10-main`.

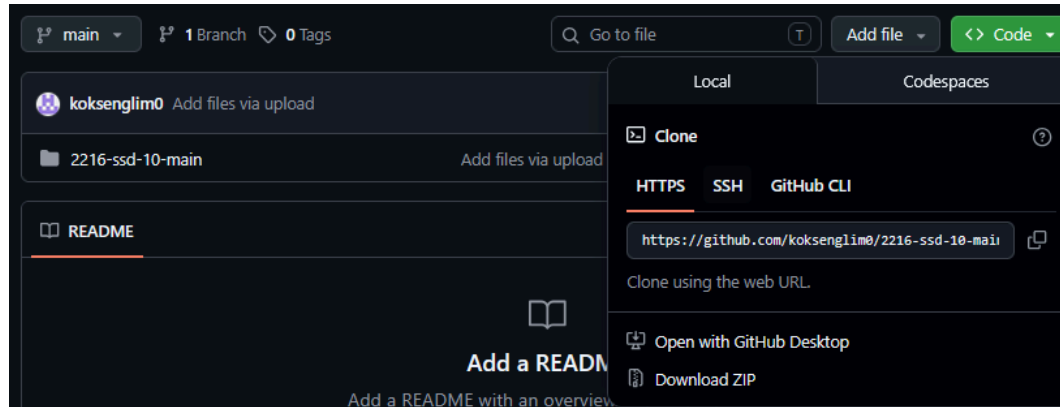
```

kokse@KOKSENG MINGW64 ~/OneDrive/Desktop/2216-ssd-10-main
$ git remote -v
fatal: not a git repository (or any of the parent directories): .git

kokse@KOKSENG MINGW64 ~/OneDrive/Desktop/2216-ssd-10-main
$ git clone https://github.com/koksenglim0/2216-ssd-10-main.git
Cloning into '2216-ssd-10-main'...
info: please complete authentication in your browser...
remote: Enumerating objects: 106, done.
remote: Counting objects: 100% (106/106), done.
remote: Compressing objects: 100% (99/99), done.
remote: Total 106 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (106/106), 124.61 KiB | 7.33 MiB/s, done.
Resolving deltas: 100% (6/6), done.

```

This meant that you will be able to get the "origin" to point to the relevant repository.
Check using `git remote -v`



Push .github into GitHub Repository

Once cloned, copy the entire `.github` you just created into the local version of repository that you just clone from GitHub and execute the following command to push the entire `.github` folder into the GitHub Repo.

```

git add .github/workflows/ci.yml
git commit -m "Add GitHub Actions CI workflow"
git push

```

Make sure the `.github` is at the repository root (that you have cloned)

```

kokse@KOKSENG MINGW64 ~/OneDrive/Desktop/2216-ssd-10-main/2216-ssd-10-main (main)
$ ls
README.md  backend/  docker-compose.local.yaml  docs/  frontend/

kokse@KOKSENG MINGW64 ~/OneDrive/Desktop/2216-ssd-10-main/2216-ssd-10-main (main)
$ git add .
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

kokse@KOKSENG MINGW64 ~/OneDrive/Desktop/2216-ssd-10-main/2216-ssd-10-main (main)
$ git commit -m "Move project files to repository root"
[main ed49e1e] Move project files to repository root
83 files changed, 91 insertions(+), 91 deletions(-)
rename {2216-ssd-10-main/.github => .github}/workflows/ci.yaml (100%)
rename {2216-ssd-10-main/.github => .github}/workflows/ci.yaml (100%)
rename 2216-ssd-10-main/README.md => README.md (100%)
rename {2216-ssd-10-main/backend => backend}/Dockerfile (91%)
rename {2216-ssd-10-main/backend => backend}/package-lock.json (100%)
rename {2216-ssd-10-main/backend => backend}/package.json (100%)
rename {2216-ssd-10-main/backend => backend}/public/app.js (100%)
rename {2216-ssd-10-main/backend => backend}/public/index.html (100%)
rename {2216-ssd-10-main/backend => backend}/public/styles.css (100%)
rename {2216-ssd-10-main/backend => backend}/scripts/check-js.js (100%)
rename {2216-ssd-10-main/backend => backend}/scripts/create-admin.js (100%)
rename {2216-ssd-10-main/backend => backend}/scripts/run-sql.js (100%)
rename {2216-ssd-10-main/backend => backend}/sql/001_schema.sql (100%)
rename {2216-ssd-10-main/backend => backend}/src/app.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/config/db.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/config/env.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/controllers/transferController.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/middleware/auth.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/middleware/csrf.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/middleware/errorHandler.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/middleware/rateLimits.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/middleware/validate.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/routes/admin.routes.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/routes/auth.routes.js (100%)
rename {2216-ssd-10-main/backend => backend}/src/routes/dashboard.routes.js (100%)

kokse@KOKSENG MINGW64 ~/OneDrive/Desktop/2216-ssd-10-main/2216-ssd-10-main (main)
$ git push origin main
Enumerating objects: 109, done.
Counting objects: 100% (109/109), done.
Delta compression using up to 16 threads
Compressing objects: 100% (95/95), done.
Writing objects: 100% (108/108), 124.47 KiB | 20.75 MiB/s, done.
Total 108 (delta 6), reused 96 (delta 6), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), done.
To https://github.com/koksenglim0/2216-ssd-10-main.git
375af22..ed49e1e  main -> main

kokse@KOKSENG MINGW64 ~/OneDrive/Desktop/2216-ssd-10-main/2216-ssd-10-main (main)
$

kokse@KOKSENG MINGW64 ~/OneDrive/Desktop/2216-ssd-10-main/2216-ssd-10-main (main)
$

```

After pushing the `.github` folder into the main, you will be able to see the workflow being executed when the triggers defined in the `ci.yaml` is triggered

The screenshot displays the GitHub Actions interface for the repository 'koksenglim0 / 2216-ssd-10-main'. The 'Actions' tab is selected, showing a workflow named 'Add GitHub Actions CI workflow with Comments (190626) #4'. The workflow is currently running, and the 'Backend CI' job is highlighted in the left sidebar. The 'Backend CI' job is shown as 'succeeded 8 minutes ago in 34s'. The job steps are listed in the main area, including 'Set up job', 'Checkout repository', 'Setup Nodejs', 'Install backend dependencies', 'Run backend audit', 'Build backend Docker image', 'Post Setup Nodejs', 'Post Checkout repository', and 'Complete job'. The 'Backend CI' job is currently running, and the 'Set up job' step is shown as 'succeeded 15s'.